

Investigating the Limitations of Desat Protection for WBG Semiconductors: Conjoint Effects due to the Parasitic Inductances, Junction Temperature and Dynamic Drain-Source Resistance

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Abstract—Due to the advent of wide bandgap (WBG) semiconductors and their adaptation into the power electronic converters, a fast and reliable overcurrent/short-circuit protection scheme is essential to ensure failsafe converter operation. Among various short-circuit protection (SCP) schemes adopted for WBG semiconductors, the desaturation (Desat) scheme is cost-effective, simple to implement with fast response times being reported in the literature. However, the impacts of parasitic inductances, which originate due to device packaging and printed circuit board layout, on the design and implementation of Desat protection are not well documented. Moreover, effect of the junction temperature of WBG semiconductors on the performance of Desat protection is not reported in the existing literature. Furthermore, the dynamic drain-source resistance characteristics, being pertinent to WBG semiconductors, pose additional challenge for reliable design of Desat protection. To ameliorate these research gaps, this paper provides a comprehensive study illustrating the conjoint effects of parasitic inductances, junction temperature and dynamic drain-source resistance on the short-circuit withstand time (SCWT) of WBG semiconductors. Thereby, this study establishes the limiting conditions imposed on the Desat protection based on the practical operating conditions and the intrinsic characteristics of WBG semiconductors. These analytical postulates are verified with the help of LTspice simulation as well as experimental results obtained based on enhancement mode GaN transistors.

Index Terms—Desaturation, dynamic drain-source resistance, overcurrent, short-circuit, wide bandgap.

I. INTRODUCTION

With the advent of emerging wide bandgap (WBG) semiconductor devices, fast and reliable overcurrent/short-circuit protection schemes have become increasingly crucial to ensure the converter reliability. WBG devices are capable of switching large currents and voltages rapidly with low losses and offer excellent physical properties, such as elevated mobility and thermal conductivity. These attributes enable smaller die size, thus reducing the junction capacitance and allowing faster switching transients [1], and lead to superior characteristics, such as higher breakdown voltage, smaller conduction losses, and minor parasitic components in comparison to Si semiconductor devices. However, the small die size of WBG devices also makes heat dissipation more challenging, particularly during short-circuit (SC) events. Furthermore, WBG

devices exhibit shorter short-circuit withstand time (SCWT) in comparison to Si counterparts. For instance, Silicon (Si) and Silicon Carbide (SiC) devices can typically withstand SC operation for up to 10 μ s [2]. On the other hand, GaN devices exhibit different SC behavior due to their unique structure and working principles. As reported in [3], for a 650 V-rated SC transistor, SC robustness is acceptable at DC voltages below 300 V, withstanding a 10 μ s SC pulse. Beyond 400 V, however, their SC withstand time drops significantly, failing within 600 ns, thus necessitating very fast SC turn-off times, thus demanding protection operation times within 200 – 300 ns for GaN HEMTs [4]. This poses stringent challenge for designing an overcurrent/SC protection scheme for WBG devices to achieve fast response as well as reliable operation [5].

One popular and widely explored technique for overcurrent/short-circuit protection is the desaturation (Desat) protection [6]–[8], which features monitoring the drain-source voltage $v_{ds(on)}$ during the ON state. When an overcurrent/short-circuit current flows into the device, the drain current i_d exceeds a certain limit, the device reaches desaturation and begins to carry the full dc bus voltage. Therefore, as the measured $v_{ds(on)}$ exceeds a preset threshold magnitude $v_{sc,th}$, a SC fault is recognized triggering the desaturation detection scheme. An advantage of this method is the direct availability of the necessary signal, i.e., the drain-source voltage v_{ds} being an easy to process signal. In addition, because of the output characteristic of unipolar devices, overcurrent/SC can be detected easily in the ohmic region of the output characteristic of device. Therefore, $v_{sc,th}$, the trigger threshold voltage during short-circuit, can be set much lower with SiC MOSFETs than with IGBTs. Furthermore, unipolar devices, such as SiC MOSFETs and GaN HEMTs, do not exhibit a ‘voltage tail’ as the bipolar devices do. Consequently, in contrast to the IGBT, the SiC MOSFET and GaN HEMT reach their steady-state voltages very steeply (for example, in less than 80 ns @ $T_c = 125^\circ\text{C}$ [9]), thus allowing fast detection of SC using Desat protection. For instance, overcurrent detection within 85 – 125 ns employing two-stage SCP scheme based on Desat technique is demonstrated in [7], [10]; however, the

tolerance for switching noise caused by the rapid dv/dt and di/dt of WBG device is not considered. Hence, necessary circuit design based on theoretical analyses of switching noise impact due to v_{ds} fluctuations is explored in [8]. This implies that designing Desat circuits for WBG devices mandates critical considerations for switching noise immunity, short-circuit detection validation and judicious design considerations.

It is apparent that selecting the threshold limit of $v_{ds(on)}$ during SC fault to generate the trip signal remains a key step for implementing the Desat protection. As high dv/dt and di/dt of WBG devices induce excessive voltage fluctuations in the v_{ds} , choosing a reliable threshold voltage $v_{sc,th}$ and defining appropriate blanking time $t_{blanking}$ for Desat protection encompassing wide range of operating conditions becomes difficult. This issue is further exacerbated due to the impacts of parasitic inductances, which originate due to the device packaging as well as the printed circuit board (PCB) layout. In addition, v_{ds} characteristic during SC condition is also influenced by the elevated junction temperature T_j of the device under fault; for instance, this study elaborates that an increase in the local ambient temperature of a traction inverter enclosure adversely affects the SCWT of device under operation and therefore demands attention to ensure reliable Desat protection under practical scenarios. Furthermore, the dynamic drain-source resistance, $D-R_{ds(on)}$, being pertinent to WBG semiconductors, pose additional challenge for the design of Desat protection. To ameliorate these research gaps, this paper provides a comprehensive study illustrating the conjoint effects of parasitic inductances, T_j and $D-R_{ds(on)}$ on the SCWT of WBG devices. First, characteristic behavior of SC faults occurring in a phase-leg configuration are discussed in Section II. Next, a LTspice simulation model is built in Section III to study the impacts of parasitic inductances and T_j during SC faults, and their implications on the Desat protection. In addition, the effect of i_{sc} is detailed in Section IV based on experimental results derived using enhancement mode GaN transistor. Finally, this paper is concluded in Section V.

II. TYPES OF SHORT-CIRCUIT FAULTS

Conventionally, SC faults are categorized into the following two categories [11]. The behavior of device under study and the relationship to the gate-drive circuitry with respect to these SC fault conditions are discussed below.

A. Hard Switch Fault (HSF)

When the device under test (DUT) is turned on, SC already exists. The test circuit shown in Fig. 1 illustrates the modes of operation for emulating HSF condition; the salient points of during this SC fault are discussed below.

a) At first, the inductive load L_{load} is short-circuited by turning on the auxiliary (AUX) device, as shown in Fig. 1(a). As the DUT is turned off, this supports the entire dc bus voltage U_{dc} , while a negative v_{gs} might be maintained.

b) Thereafter, the DUT is turned on directly into the SC fault. The rate at which the DUT begins to conduct current and the magnitude of the fault current are related to the charging

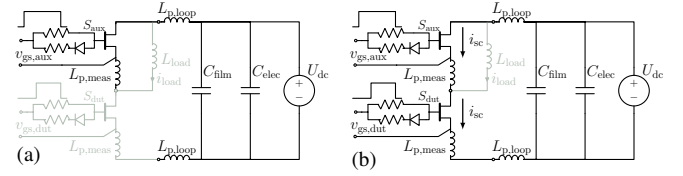


Fig. 1: Test circuit stages of operation during HSF emulation (a) the AUX device is turned on at first, followed by (b) the inadvertent turn on of the DUT, establishing short-circuit in the phase-leg.

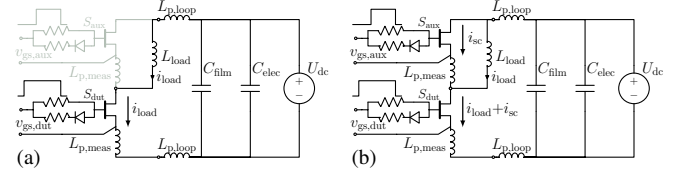


Fig. 2: Test circuit stages of operation during FUL emulation (a) the DUT is turn on to build the load current, which is followed by (b) inadvertent turn on of the AUX device applying short-circuit in the phase-leg.

rate of the input capacitance and v_{gs} . Upon application of the gate signal and following the v_{gs} reaching its threshold voltage, the drain current i_d begins to rise. Effectively, the fault current i_{sc} is limited by the device saturation, biased according to v_{gs} . The device is able to withstand the simultaneous load of a high current and a high voltage during the short-circuit pulse for some time. However, it must be turned off within a defined time, typically specified by the device manufacturer, to ensure safe operation and to avoid failures due to overheating.

c) A notch is present in v_{ds} due to the voltage drops present across the resistive and inductive elements of the circuit. The current i_{sc} decreases with time due to device self-heating.

d) During turn-off of i_{sc} , the rapid fall of the fault current combined with the loop inductance causes v_{ds} to overshoot the rated dc bus voltage U_{dc} . The condition for surviving of a short-circuit pulse is that the peak voltage must remain lower than the rated voltage within the specified safe operating area (SOA). To ensure this, short-circuit must be turned off with a limited di/dt . The rate of fall of the fault current and hence the magnitude of overshoot of v_{ds} can be controlled to a certain degree, by using higher gate resistor $R_{g,off}$ leading to a soft turn-off. However, increasing $R_{g,off}$ to reduce the overvoltage transient during HSF is in conflict with other design considerations, such as the gate noise immunity and the dv/dt capability [12], and therefore requires a trade off.

B. Fault Under Load (FUL)

A device may be subjected to a SC fault while operating under normal conduction. The test circuit shown in Fig. 2 indicates the stages of operation for emulating FUL condition; the key observations are noted below.

a) Initially the DUT is gated on and is operating normally, carrying a steady i_{load} within its ratings [cf. Fig. 2(a)]; the drain-source voltage of the DUT is low and equal to $v_{ds(sat)}$.

b) As soon as the SC fault is imposed on the device by turning the AUX device on, causing a shoot-through as shown in Fig. 2(b), i_d increases very steeply, pulling the device out of its near saturation state. The di/dt is determined

by U_{dc} and the inductance of the short-circuit loop. The device being desaturated, the consequently high dv/dt of v_{ds} induces a displacement current through the gate-drain (Miller) capacitance C_{gd} , causing v_{gs} to increase to a higher level. This increase of v_{gs} in turn causes a high i_{sc} peak.

c) Since C_{gs} is high at low v_{ds} , effect of the displacement current is most significant at low voltage. The increased v_{gs} during desaturation leads to a negative gate current i_g back to the gate driver. The onset of negative i_g together with the interconnection parasitics and device input capacitances cause oscillations in gate-source voltage v_{gs} .

d) After the completion of desaturation phase, i_{sc} drops to its static value. Due to the negative di/dt , voltage drop induced over the parasitic inductance imposes a voltage peak on v_{ds} .

e) The stationary phase is followed by the turn-off of i_{sc} , usually achieved by withdrawing the gate signal. Due to the negative di/dt , the commutation circuit inductance again induces an overvoltage across the device. Therefore, it must be ensured that these voltage peaks stay below the rated voltage of the device, within the specified safe operating area (SOA).

f) To limit the peak fault current, hence resulting in lower power dissipation, and to keep gate-source voltage within the permissible limits, v_{gs} needs to be clamped. This can be achieved by reducing the value of $R_{g,off}$. This facilitates better clamping of v_{gs} , thereby reducing the fault current magnitude significantly [11]. This improves the SCWT of device, allowing larger response time available for the protection circuit.

During HSF, only a small positive dv/dt is generated across the device. Thus the Miller effect, a major consideration during FUL, has much less influence in case of HSF, particularly because the Miller capacitance is much lower at higher voltage. Therefore, HSF can result in much lower fault current than FUL. In this way, a low inductive FUL condition is more severe in comparison to HSF. Being less stressful, HSF test tends to yield optimistic results in terms of the short-circuit capabilities of a device [11]. That is why, the parametric sweep analysis in Section III-D is based on FUL condition.

III. EFFECTS OF PARASITIC INDUCTANCES AND JUNCTION TEMPERATURE ON THE DESAT PROTECTION

A. LTspice Model

In order to study the effects of parasitic inductances and junction temperature of device on the SC faults discussed in the previous section, the simulation model shown in Fig. 3 is built and emulated in LTspice. The relevant design parameters are summarized in Table I. One phase-leg configuration is modeled based on the enhancement mode GaN transistor GS66516T by GaN Systems [13]; the GaN_LTspice_GS66516T_L3V4P1 model provided by GaN Systems is used which integrates the thermal model of device and provides access to the respective junction and case temperatures of the device; details of the thermal resistance calculation from case to heatsink and then to ambient, and other parameters mentioned in the table. Based on these simulation parameters, the characteristics of $v_{ds(on)}$ are studied during both normal turn on as well as different SC fault

TABLE I: Simulation parameters to study SC faults in LTspice

Parameters	Values
DC bus voltage (U_{dc}) ¹	400 V
Decoupling capacitors (C_{elec} , C_{cera}) ²	200 μ F, 2.1 μ F
Gate resistances ($R_{g,on}$, $R_{g,off}$) ³	10 Ω , 1 Ω
Parasitic inductances ($L_{p,meas}$, $L_{p,loop}$)	1 nH, 1 – 4 nH
Thermal resistance ($R_{th,ca}$) ⁴	0.67 Ω
Load inductance (L_{load}) ⁵	45 – 112.5 mH

¹ Stray dc bus inductance L_{bus} is assumed to be 1 nH.

² The equivalent series resistance (ESR) and equivalent series inductance (ESL) of the electrolytic and ceramic dc bus capacitors are $ESR_{elec} = 281$ m Ω , $ESL_{elec} = 5$ nH, and $ESR_{cera} = 3$ m Ω , $ESL_{cera} = 5$ pH.

³ Stray inductance of the gate driver circuit $L_{g,ext} = 5$ nH is assumed.

⁴ The thermal resistance from case to ambient $R_{th,ca}$ is calculated based a nominal airflow rate of 0.5 m/s and natural convection process is assumed.

⁵ Load inductance of 67.5 mH corresponds to a load current of $i_{load} = 23.6$ A at the time instant of initiating SC fault in the phase-leg.

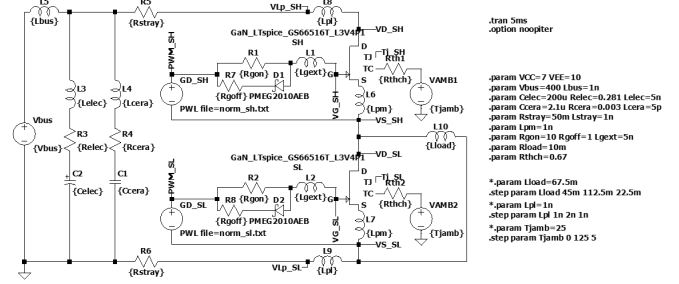


Fig. 3: Simulation model built in LTspice to study the effects of parasitic inductances and junction temperature on the short-circuit withstand time of GaN transistor GS66516T from GaN Systems [13].

conditions; thereby, the effects of the parasitic inductances and junction temperature are studied individually and their impacts on the choice of threshold voltage $v_{sc,th}$, required for implementing the Desat protection, are quantified next.

B. Effect of Power Loop Inductance $L_{p,loop}$ on $v_{ds(on)}$

1) *Normal Turn ON*: The simulation results obtained during normal turn on operation are shown Fig. 4 illustrating the effect of different parasitic inductance components, i.e., $L_{p,meas}$ and $L_{p,loop}$. The scope parameters shown in dark and gray colors indicate the drain-source voltages measured without and with additional loop inductance, respectively; the threshold voltage limit $v_{sc,th} = 3$ V is assumed based on literature [7]. It is apparent that presence of stray inductance give rise to the unwanted oscillation in the measured drain-source voltage $v_{ds(on)}$; both the overshoot magnitude and settling time of these natural oscillations increase with larger parasitic inductances, which demand introducing higher blanking time $t_{blanking}$. For instance, Fig. 4 depicts that with introduction of only 1 nH of additional $L_{p,loop}$, blanking time $t_{blanking}$ needs to be increased by 226.5% to avoid false triggering during normal operation; this makes difficult to choose a suitable $t_{blanking}$ necessary for implementing the Desat protection. Moreover, as $t_{blanking}$ is increased to counter the effect of higher oscillation due to parasitics, this inhibits fast/immediate detection of i_{sc} using Desat protection and allows the fault current to increase until $t_{blanking}$ is expired, leading to higher current stress, loss and temperature rise in the device.

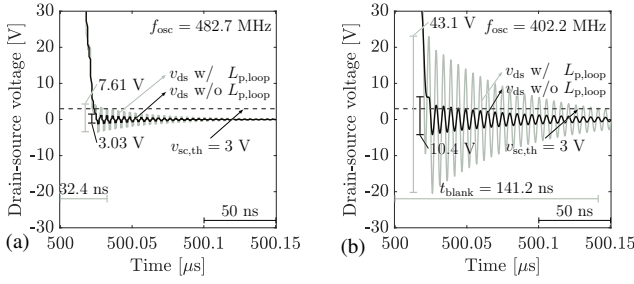


Fig. 4: Effect of the parasitic loop inductance $L_{p,loop}$ on the drain-source voltage v_{ds} during normal operation (a) without any $L_{p,loop}$, and (b) with additional $L_{p,loop} = 1$ nH (device under study is the high-side GaN, $I_{load} = 23.6$ A; under both test conditions, $L_{p,meas} = 1$ nH. $T_j = 25$ °C).

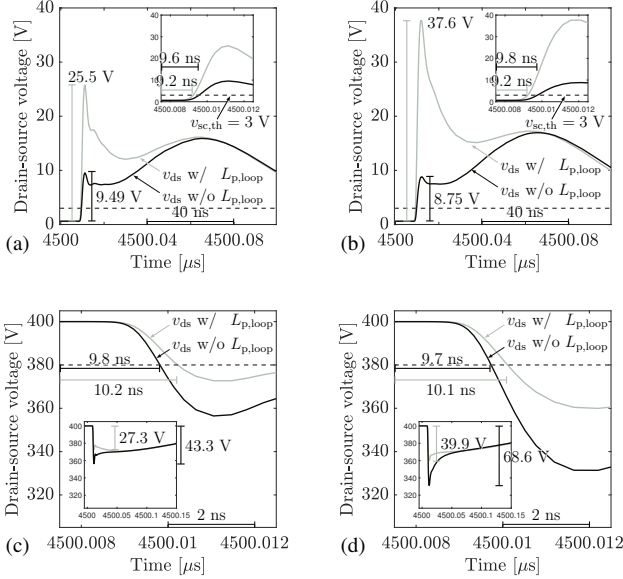


Fig. 5: Effect of the loop inductance $L_{p,loop}$ on $v_{ds(on)}$ during (a)–(b) HSF, and (c)–(d) FUL, respectively. Traces are shown in (a)–(c) without $L_{p,loop}$, whereas in (b)–(d) an additional $L_{p,loop} = 1$ nH is considered. (for all scenarios, $I_{load} = 23.6$ A; $L_{p,meas} = 1$ nH and $T_j = 25$ °C; the drain-source voltage $v_{ds(on)}$ is measured across the high-side GaN).

2) **Hard Switch Fault (HSF):** Results obtained under HSF are shown in Figs. 5(a)–(b), for without $L_{p,loop}$ and with an additional $L_{p,loop} = 1$ nH in the power loop, respectively. Characteristic of $v_{ds(on)}$ during HSF is very different from that observed during normal turn ON. The results indicate that given the rise time of $v_{ds(on)}$ during fault condition is very fast, the effect of additional $L_{p,loop}$ remains practically same except for higher overshoot amplitude of $v_{ds(on)}$; once the initial transients die out, both traces practically converge. Although the measured $v_{ds(on)}$ during fault condition exceeds that of $v_{sc,th}$ ($= 3$ V [7]) within a very short time period (≈ 10 ns), the Desat protection can only set in after $t_{blanking}$ is expired and highlights the inherent constraint of Desat protection.

3) **Fault Under Load (FUL):** Similar results derived under FUL are shown in Figs. 5(c)–(d), respectively. In this case, an additional $L_{p,loop} = 1$ nH reduces the fall time of $v_{ds(on)}$ by 9.8% and increases its undershoot amplitude by 62.5%. This concludes that higher $L_{p,loop}$ increases the amplitude change in $v_{ds(on)}$ during FUL and therefore impacts the sensitivity and

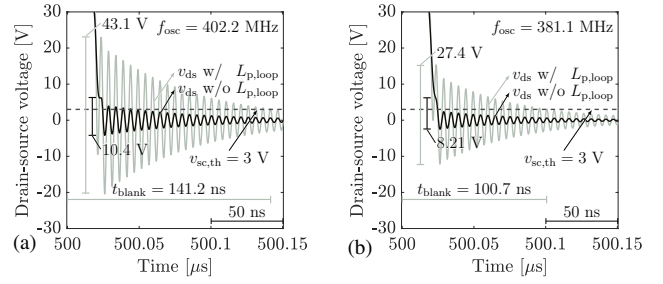


Fig. 6: Effect of the junction temperature T_j of GaN device on v_{ds} during normal operation while the junction temperature is maintained at (a) $T_j = 25$ °C, and (b) $T_j = 75$ °C, respectively (device under study is the high-side GaN, for both cases, $I_{load} = 23.6$ A; $L_{p,meas} = 1$ nH and $L_{p,loop} = 1$ nH).

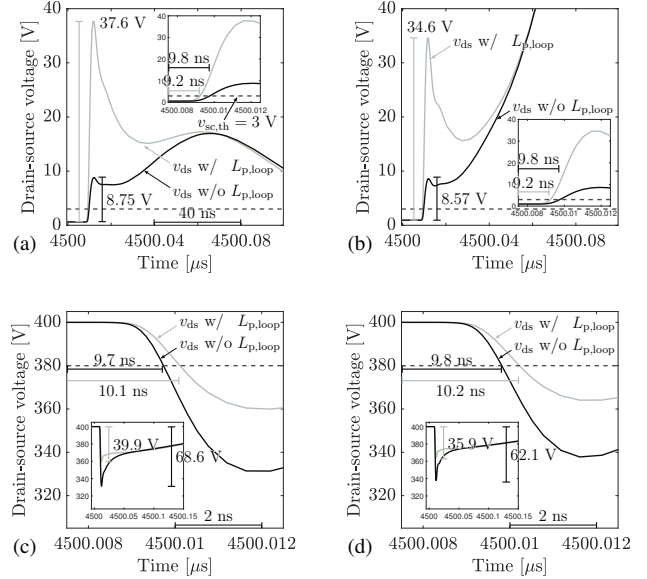


Fig. 7: Effect of the junction temperature T_j on $v_{ds(on)}$ during (a)–(b) HSF, and (c)–(d) FUL, respectively. Traces are shown for T_j at (a)–(c) 25 °C, and (b)–(d) 75 °C, respectively. (for all cases, $I_{load} = 23.6$ A; $L_{p,meas} = 1$ nH and $L_{p,loop} = 1$ nH; the drain-source voltage $v_{ds(on)}$ is measured across the high-side GaN).

design of the sensing circuit employed for Desat protection. This effect can be exacerbated further if the sag/swell of U_{dc} is taken into account which may take place during transient conditions in case of a traction inverter.

C. Effect of Junction Temperature T_j on $v_{ds(on)}$

1) **Normal Turn ON:** Based on the device and heatsink thermal models discussed in Section III-A, the effect of device T_j is studied. Simulation results obtained during normal ON are shown in Fig. 6 for two different T_j s. Due to the positive temperature coefficient α , the drain-source on resistance $R_{ds(on)}$ increases at higher T_j ; this introduces higher damping, and thereby reduces the settling time of the natural oscillations in $v_{ds(on)}$. Therefore, for a given $v_{sc,th}$, the selection of $t_{blanking}$ is dictated by the lower limit of T_j of the device. For instance, a relative decrement of 200% in T_j leads to an increment of 53.6% in the required $t_{blanking}$. Therefore, T_j of device, being governed by either the local ambient and/or load, as is the case of a traction/photovoltaic inverter, should also to be weighed in carefully while implementing Desat protection.

2) *Hard Switch Fault (HSF)*: Results are shown in Figs. 7(a)–(b) for $T_j = 25^\circ\text{C}$ and 75°C , respectively. It depicts that the time taken for $v_{ds(on)}$ to reach the threshold limit of $v_{sc,th}$ remains practically same for both cases; the overshoot amplitude is reduced marginally but the rate of rise of $v_{ds(on)}$ is exceedingly large at higher T_j . For instance, at $T_j = 25^\circ\text{C}$, following an overshoot of 37.6 V $v_{ds(on)}$ gradually decreases to low value. However, in case of $T_j = 75^\circ\text{C}$, although the first voltage overshoot is 7.98% less, $v_{ds(on)}$ decrease initially and then continues to increase monotonically at a steep rate of 721 V/ μs , implicating higher loss across the device, and a possible thermal runaway. This illustrates that SCWT of GaN device is adversely affected at higher junction temperature.

3) *Fault Under Load (FUL)*: Results shown in Figs. 7(c)–(d) for FUL indicate that while time taken to reach the threshold limit remains practically same, the undershoot amplitude of $v_{ds(on)}$ decreases marginally at higher T_j . For instance, 10% decrement in $\Delta v_{ds(on)}$ is noted due to 200% increment in T_j .

D. Parametric Sweep Analysis

To analyze the conjoint effects of parasitic loop inductances and junction temperature on the short-circuit withstand time of GaN device, this paper conducts a parametric sweep analysis based on the LTspice model shown in Fig. 3. The methodology illustrated in Fig. 8 indicates that three design parameters are swept within the following limits: the parasitic loop inductance in the power loop $L_{p,loop} \in (0, 1, 4)$ nH, ambient temperature $T_{amb} \in (0 - 125)^\circ\text{C}$ in steps of 5°C , and load inductance $L_{load} \in (45, 67.5, 90, 112.2)$ mH to vary the load current i_{load} in steps. For each combination, the SCWT is calculated as time required for T_j of the device to reach 150°C from the instant SC fault is initiated. In this study, FUL condition is emulated as this is more severe in comparison to HSF. The resulting SCWT distribution data as obtained are shown in Fig. 9. Based on the sweep analysis, the following findings are noted: while the remaining circuit parameters are kept constant, SCWT

- decreases monotonically as i_{load} increases (for example, in case of the operating point at $L_{p,loop} = 5$ nH and $T_j = 95^\circ\text{C}$, SCWT of upper GaN device decreases by 35.2% while i_{load} is increased by 50% (from 23.6 A to 35.4 A),
- decreases with $L_{p,loop}$ (for instance, in case of the operating point at $i_{load} = 35.4$ A and $T_j = 105^\circ\text{C}$, SCWT is decreased by 10.5% while the total parasitic inductances (sum total of $L_{p,meas}$ and $L_{p,loop}$) is increased from 1 nH to 5 nH).
- reduces as T_j increases (e.g., at the operating point $i_{load} = 35.4$ A and $L_{p,loop} = 4$ nH, SCWT of high-side GaN is decreased by 74.5% as T_j is increased from 25°C to 115°C).
- the limitation of SCWT is imposed on both upper and lower GaN devices at different load conditions, as shown in Fig. 9. These results indicate that the SCWT of device under protection is concurrently dependent on several design parameters, such as i_{load} , parasitic inductances and T_j . Therefore, their conjoint effects should be taken into account to improve the reliability of Desat protection under wide range of operating conditions in terms of output load, ambient temperature, and parasitic inductances associated with device and PCB traces.

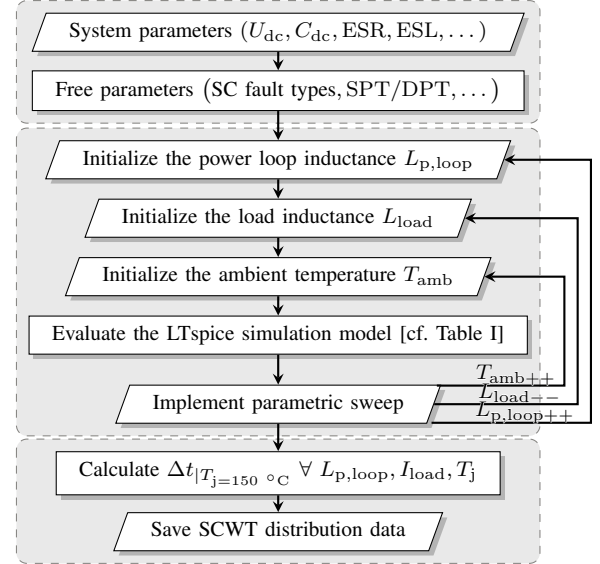


Fig. 8: Framework of the parametric sweep analysis to calculate the short-circuit withstand time (SCWT) of GaN device as the load current I_{load} , junction temperature T_j and parasitic inductance $L_{p,loop}$ are varied in steps.

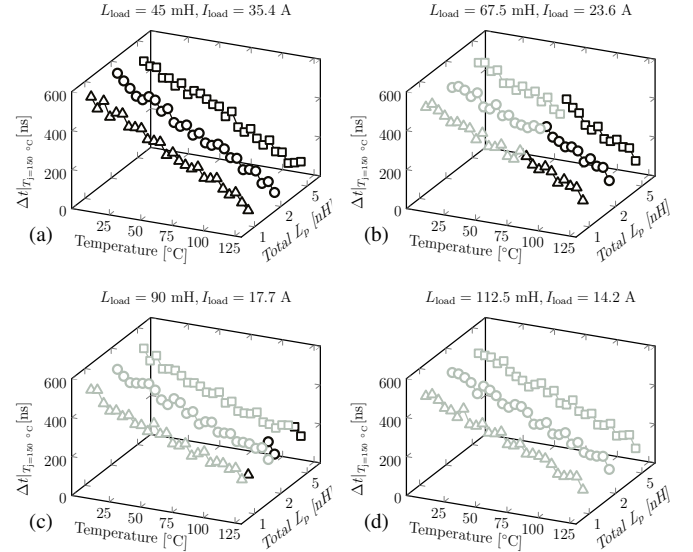


Fig. 9: Results of the parametric sweep analysis illustrating the nature of SCWT of GaN device under the influence of conjoint effects due to load current I_{load} , junction temperature T_j and stray inductance $L_{p,loop}$ (d)–(d) load current is reduced in steps from 1 p.u. to 0.25 p.u.; markers shown in black and gray colors indicate the SCWT for the upper and lower GaN devices, respectively, at different junction temperature and stray inductances. SCWT is defined as the time duration taken for T_j to reach 150°C under sustained short-circuit fault condition ($L_{p,meas} = 1$ nH is assumed; the parameter L_p is the sum total of parasitic inductances $L_{p,meas}$ and $L_{p,loop}$).

IV. EXPERIMENTAL RESULTS: EFFECTS OF DYNAMIC DRAIN-SOURCE RESISTANCE ON THE DESAT PROTECTION

It is apparent from the previous discussion that designing the Desat protection for WBG semiconductors is challenging due to their lower SC tolerance and mutual dependency on the parasitic inductances and the device junction temperature. This mandates that the threshold value $v_{sc,th}$ must be carefully designed: if set too high, faults may go undetected or lead to

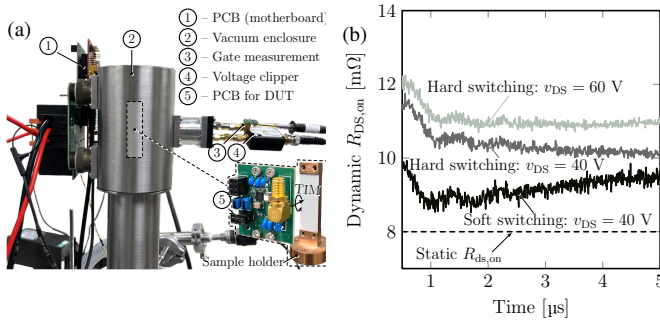


Fig. 10: (a) An automated characterization platform for comprehensively characterizing the dynamic $R_{ds(on)}$ of GaN devices over wide temperature ranging from 40 K to 400 K (detailed description can be found in [14]). (b) The measured waveform illustrate the dependency of dynamic $R_{ds(on)}$ on the circuit operations, such as hard/soft-switching operation and the dc bus voltage. A double pulse test (DPT) circuit (similar to what is illustrated in Fig. 1) is employed as the testbench.

delayed protection; if too low, false triggers can occur.

However, the above-said criterion is rendered even more difficult due to an additional challenge: the dynamic effect, a phenomenon unique to lateral GaN devices [15]. At turn-on, the actual $R_{ds(on)}$ is significantly higher than its static value, leading to an elevated $v_{ds(on)}$. Therefore, the SC threshold cannot be designed too small, otherwise it will increase the risk of mis-trigger. Moreover, dynamic characterization is not usually covered in the datasheets or simulation models provided by the vendor, which makes it hard to evaluate $v_{ds(on)}$ before the test, thus further complicating the SC threshold design. In addition, dynamic $R_{ds(on)}$ varies with various factors like switching mode, voltage, current, gate resistance, temperature, and frequency [16]. Experimental results derived based on dynamic characterization are shown in Fig. 10(b) illustrating the nature of dynamic $R_{ds(on)}$ characteristics of EPC2307 GaN device [17], measured at various levels of dc bus voltages and switching conditions, such as hard-/soft-switching operations. The measured results indicate that the dynamic increases as the dc bus voltage U_{dc} is increased, thus making it difficult to establish a generic short-circuit threshold design for $v_{sc,th}$, which is required to successfully derive a Desat protection scheme for SC fault detection.

V. CONCLUSIONS

This paper studies the conjoint effects of parasitic inductances, junction temperature and dynamic drain-source resistance on the short-circuit withstand time (SCWT) of WBG devices and therefore highlights the limitations of conventional Desat protection. The key findings are summarized as: SCWT of WBG semiconductor *a)* decreases monotonically as i_{load} increases, *b)* reduces as the sum total of parasitic inductances $L_{p,meas}$ and $L_{p,loop}$ increase, and *c)* decrease as T_j of device increases, which may occur due to higher ambient and/or increased load. These conjoint effects render implementing Desat protection difficult under practical operating scenarios. For instance, with introduction of only 1 nH of additional $L_{p,loop}$ requires increasing $t_{blanking}$ by 226.5% to avoid false triggering during normal operation. In addition, for a given

$v_{sc,th}$, a relative decrement of 200% in T_j leads to an increment of 53.6% in the required $t_{blanking}$. Therefore, while selecting $t_{blanking}$ for Desat protection, higher limit of the parasitic inductance should be considered, while lower limit of the device junction temperature need to be taken into account. Furthermore, characteristic of dynamic drain-source resistance $D-R_{ds(on)}$ being dependent on the switching mode, dc bus voltage, etc., render it difficult to establish a generic short-circuit threshold design for conventional Desat protection, and thus warrants further investigations to improve the reliability of fast-response SCP schemes for detecting overcurrent/short-circuit of high power WBG semiconductor devices.

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